Message-Passing GNN for Antibiotic Discovery

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***Abstract*—Our project is based on the 2023 Nature paper *'Deep Learning-guided discovery of an antibiotic targeting Acinetobacter baumannii'* [1]. We identified this as a paper that made several architectural claims which were left unexplained. It also only ran inference on a very small dataset without much chemical diversity. We reimplemented their work; varied their choice of ensemble size, dropout and message-passing depth; and ran inference on a far larger, more chemically diverse dataset. We achieved increased AUROC when using dropout (not used in the paper), noticed negligible differences across values for message-passing depth, and found the ensemble approach to be warranted. Additionally, running inference on the larger COCONUT marine products dataset [7] led to insights into the model's lack of ability to generalize to molecules that are outside of its very limited training set. We discovered that this paper's main bottleneck was its small training set, limited by the necessity of lab time in producing these labeled data points.**

GitHub Repository:
https://github.com/theodore-casares/chemprop-redux

I. Introduction

We implemented *'Deep learning-guided discovery of an antibiotic targeting Acinetobacter baumannii'* [1] from Liu et al. This paper trains a message passing neural network to predict which molecules may display activity against A. baumannii. A. baumannii is a bacterium that causes severe healthcare associated infections.

We then ran their model on the same data to test their quantitative claims on the number of molecules the model predicts as being valid antibiotics, as well as their AUROC on the test set.

Additionally, we perform three ablation studies. We investigate whether the paper's use of an ensemble of ten models, a message passing depth of three, and zero dropout are warranted. We will view these decisions through the frame of AUROC, AUPRC, and accuracy. We identified these tasks as they provide insight into which architectural decisions were most consequential. We also run the original model on the COCONUT marine products dataset [7]. This was not investigated in the original paper, and was done to investigate the model's ability to generalize to molecules with chemical groups not seen in the training data.

A. Motivation

The cost to develop a drug in the U.S. has risen aggressively in recent decades. [2] Traditional high-throughput screening can test millions of molecules and cost millions of dollars. This work effectively pre-selects which molecules are worth being screened, saving significant time and money. The Liu et al. paper shows this in action, and even details the pipeline from developing the model to testing the model's top predictions in a lab.

We were also inspired by how active deep learning research in drug discovery is. Projects like DeepMind's AlphaFold2 are extremely exciting, with Demis Hassabis and John Jumper winning the 2024 Nobel Prize in Chemistry. [3] During CSCI 1470, we only had one lecture on Graph Neural Networks (GNNs). This project also enabled us to gain some experience in working with GNNs and understanding one of their most powerful applications - drug discovery.

B. Goals

To explicitly show how we are aligned with the tasks discussed in the handout, we introduce our quantitative claims and hypothesis-driven ablations here.

- **Base Goal:** Reimplement the original model from Liu et al. [1] and investigate their quantitative claims. Run inference on the COCONUT marine products dataset [7] with this original model.
- **Target Goal:** Alter the paper's architecture by changing the message-passing depth, ensemble size, and by introducing dropout. Provide a clear analysis of these results.
- **Stretch Goal:** Train multiple ensembles with each of the values for dropout, ensemble size, and message-passing depth that we are investigating. This is in hopes of limiting noise from random initialization in our results. Run inference on the COCONUT marine products dataset [7] using these ensembles.
- **Quantitative Claims:**
  1) The authors claim that their ensemble produces 240 valid molecules after filtering and running inference on the Drug Repurposing Hub.
  2) The authors claim that their ensemble model has an AUROC of 0.792 ± 0.042.
- **Hypothesis-Driven Ablations:**
  1) The paper uses a message-passing depth of three. This is not justified by the authors and may lead to a loss of the local chemical distinctions.
  2) The paper adopts an ensemble of ten models. We believe this may be unnecessary due to how well our first individual model performed.
  3) The paper does not use dropout in its model. We believe this may be unreasonable. They have a very

small training set, and introducing dropout may prevent overfitting.

II. Literature Review

Our work builds on 'Deep learning-guided discovery of an antibiotic targeting Acinetobacter baumannii' (Nature, 2023) by Liu et al. [1].

- **Summary of key papers:** In 2019, Yang et al. released 'Analyzing Learned Molecular Representations for Property Prediction' [4]. This introduced Chemprop, a PyTorch-based framework for training and evaluating message-passing neural networks. Liu et al. [1] use this as the backbone of their paper, where they train a message-passing graph neural network (MPGNN) to predict which molecules will inhibit the growth of A. baumannii. They then tested the top predictions in a lab setting, leading to the discovery of abaucin, a narrow-spectrum antibiotic.
- **Existing landscape:** Liu et al. [1] use an MPGNN with approximately 400,000 parameters. They train it on a labeled dataset of approximately 7,700 molecules. The training set is quite small due to the fact that labeled data is expensive and time-consuming to garner when lab work is required to produce it. They use a ten-model ensemble without dropout and a message-passing depth of 3. Using this model, they then run inference on the approximately 6,000 molecules from the Drug Repurposing Hub [1].
- **Gaps in prior research:** Liu et al. do very little in the way of justifying their architectural choices. For example, there is no mention of why zero dropout is used, or why a message-passing depth of 3 is adopted. Additionally, the Drug Repurposing Hub is a database that only contains FDA-approved drugs [5]. They do not investigate how the model handles compounds that may kill A. baumannii, but are also dangerous to humans.
- **How our project is an extension of existing work:** We stress-test several architectural choices from the Liu et al. paper. We investigate the effects of using a single model over an ensemble. We also see how introducing dropout and varying the message-passing depth affect the ensemble's performance. Additionally, we clean (remove leakage) and run inference on the COCONUT marine products dataset [7]. This contains over half a million molecules, some of which are dangerous. We provide insight into the model's ability to generalize to chemical groups it does not see in training. We also illustrate how the model handles molecules that are not feasible for use as drugs.

III. Methodology

A. Dataset

We use three different datasets across training and evaluation:

- **Source and size:** Training uses SD1 from Liu et al. [1], containing 7,686 molecules, all screened and labeled against *A. baumannii* with lab-verified continuous growth inhibition scores. For evaluation, we used two different datasets: SD2 (the Drug Repurposing Hub [5], 6,647 raw molecules) for direct comparison with the original paper, and COCONUT (Collection of Open Natural Products, 738,827 raw marine natural products) [7] as a chemistry stress-test not seen in the original paper.
- **Preprocessing steps:** SD1's continuous scores are converted to binary at $\mu - \sigma$, following the original paper. This yields 480 actives and 7,204 inactives (6.25% positive rate). All SMILES strings are canonicalized using the RDKit library. SD2 is canonicalized, deduplicated, and stripped of any molecule whose canonical SMILES exactly matches an SD1 entry, leaving 5,864 usable evaluation molecules. COCONUT [7] is filtered to molecules with at most one Lipinski-rule violation, deduplicated, and stripped of SD1 overlap, leaving 502,874 valid evaluation molecules.
- **Train/validation/test splits:** SD1 is split 80/10/10 randomly per seed, with a fresh split drawn from each of the ten ensemble seeds. SD2 and COCONUT [7] are evaluation-only, lacking the continuous growth inhibition labels of SD1, due to a need for physical in-lab testing. They are both scored without splitting.
- **Data augmentation:** None. Molecular graphs are fully deterministic functions of the canonical chemical representation of SMILES, and the 200-dimensional `rdkit_2d_normalized` global featurization is only computed once per molecule.

B. Model Architecture

We reimplement the directed message-passing neural network (D-MPNN) of Yang et al. [4], which forms the backbone of the Liu et al. pipeline. Each molecule is represented as a graph in which atoms are nodes and bonds are edges. Atom features (133 dimensions) encode an atom's atomic number, degree, charge, chirality, hybridization, aromaticity, hydrogen count, and mass. Bond features (14 dimensions) encode bond type, conjugation, ring membership, and stereochemistry.

Unlike standard message passing networks that propagate messages between atoms, the D-MPNN propagates messages along directed bonds. This avoids the totters problem in which information passes redundantly back and forth across the same edge. For each directed bond (v, w), the initial hidden state $\mathbf{h}_{vw}^{(0)}$ is a learned linear transform of the concatenated atom features of v and the bond features of (v, w). Over K iterations, the hidden state gets updated by aggregating messages from all of the incoming directed bonds at v, except for the reverse bond (w, v):

$$\mathbf{h}_{vw}^{(\ell)} = \mathrm{ReLU}\left(\mathbf{h}_{vw}^{(0)} + \mathbf{W}_h \sum_{u \in N(v) \setminus w} \mathbf{h}_{uv}^{(\ell-1)}\right) \tag{1}$$

where $\mathbf{W}_h \in \mathbb{R}^{d \times d}$ is the shared weight matrix across iterations and $N(v)$ denotes the neighbors of atom v.

After K rounds of message passing, the atomic-level hidden states are recovered by aggregating the incoming bond messages and combining them with the atom's original features:

$$\mathbf{h}_v = \mathrm{ReLU}\left(\mathbf{W}_o \left[\mathbf{x}_v \,;\, \sum_{u \in N(v)} \mathbf{h}_{uv}^{(K)}\right]\right) \tag{2}$$

where $\mathbf{x}_v$ represents the original feature vector of atom v and $[\cdot\,;\,\cdot]$ denotes concatenation. A graph-level molecular representation is produced by summing atom hidden states across the molecule.

This learned representation is then concatenated with 200 RDKit-derived global molecular descriptors, which provide chemistry-specific signals.

$$\hat{\mathbf{y}} = \mathrm{sigmoid}\Big(\mathbf{W}^{(2)}\,\mathrm{ReLU}\Big(\mathbf{W}^{(1)}\mathbf{h}^{\mathrm{mol}} + \mathbf{b}^{(1)}\Big) + \mathbf{b}^{(2)}\Big) \tag{3}$$

yielding the predicted probability that the molecule inhibits *A. baumannii* growth. Total parameter count is approximately 400,000.

Our implementation's baseline matches the hyperparameters used by Liu et al. [1]: message passing depth $K = 3$, hidden size 300, two FFN layers, and dropout 0. Ablation runs vary these axes.

C. Training Procedure

- **Loss function:** Binary cross entropy with logits over the SD1 training split:

$$\mathcal{L}(\hat{\mathbf{y}}, \mathbf{y}) = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log \hat{y}_i + (1 - y_i)\log(1 - \hat{y}_i)\right] \tag{4}$$

  where $y_i \in \{0, 1\}$ is the binarized activity label and $\hat{y}_i$ is the model's predicted probability.
- **Optimizer (e.g., Adam, SGD) and learning rate:** Adam with a Noam learning-rate schedule, matching the Chemprop default used by Liu et al.
- **Batch size and number of epochs:** Batch size 50, trained for 30-20 epochs per model (we found that 30 was overkill, with AUROC usually peaking below 10 epochs). The best-validation-AUC checkpoint is saved for every seed.
- **Regularization techniques:** The baseline implementation uses no dropout, following Liu et al. Ablations introduce dropout $\in \{0.1, 0.2, 0.3\}$ in the FFN head and message-passing layers. We also train 10-model ensembles (one model per seed, seeds 0-9), with each seed producing an independent 80/10/10 split and random initialization. The ensemble predictions are the unweighted mean of the per-model sigmoid outputs.
- **Hyperparameter tuning approach:** Instead of tuning for absolute performance, we sweep three different architectural axes against the paper's reported configuration: dropout $\in \{0, 0.1, 0.2, 0.3\}$, message-passing depth $K \in \{1, 3, 5\}$, and ensemble size (1 vs. 10). Each configuration is trained as a full 10-seed ensemble to control for split variance.

D. Evaluation Metrics

The paper uses a few metrics to measure the robustness of their model. Area Under the Receiver Operating Characteristic Curve (AUROC) can be understood as the model's ability to distinguish between two classes, plotting the true positive rate against the false positive rate. The Area Under the Precision-Recall Curve (AUPRC) is more applicable when the dataset is highly imbalanced, such as the original paper's. AUPRC focuses on precision and recall: the model's ability to find all the positive cases without making too many errors. Accuracy is simply the proportion of correct predictions, but it is not as useful in this application since guessing "inactive" on every molecule would still result in an accuracy of 93.75%.

IV. Results

A. Novel Dataset Application

The filtered COCONUT dataset scored 502,874 unique molecules for effectiveness in inhibiting growth against A. baumannii. Notably, many molecules received a perfect 1.0 score. For example, molecules including a diazo/diazonium group produced a 1.0 score. However, they are generally considered toxic and are potentially explosive [6]. This group is not in the training set, and the model is effectively performing an out-of-distribution prediction.

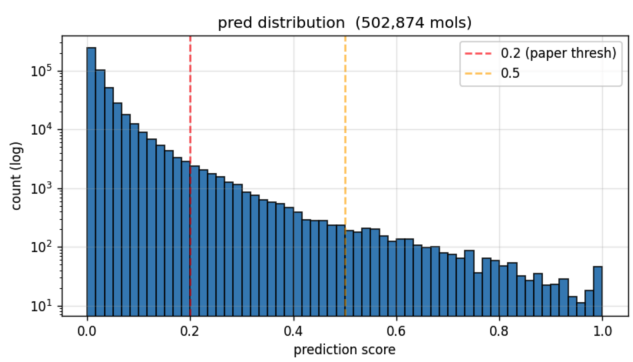

Fig. 1: COCONUT Prediction Scores

The first quantitative claim we tested was the original paper's reported train AUROC and AUPRC. After running a single model trained and tested on seed 0, we achieved an AUROC and AUPRC within one standard deviation of the paper's reported numbers. Thus, we concluded the paper's AUROC and AUPRC are reasonable.

Secondly, we tested whether applying our model to SD2 and filtering by raw score and Tanimoto similarity would yield the same number of priority molecules as the original paper: 240. After running SD2 through our model and removing SD1 duplicates, we found 545 molecules with a prediction score above 0.2. After further filtering out molecules with a Tanimoto similarity score above 0.3, we were left with 240 molecules: the exact same number as the paper.

B. Ablations

Initially, we believed that a 10-model ensemble was unnecessary. After all, we achieved an AUROC of 0.825 and AUPRC of 0.380 from our first run on seed-0. Both of these numbers were higher than the paper's mean, so we figured the ensemble wasn't needed. However, when performance across dropout is averaged by seed, we see the full picture. Since the dataset is so small, the train/test split makes a big difference in the metrics. These differences are misleading: one seed may train on more relevant molecules and test on easier molecules, and vice versa. Our one-seed model was trained and tested on seed 0, second only to seed 4 in average performance. Such high variance between models suggests that an ensemble approach is logical.

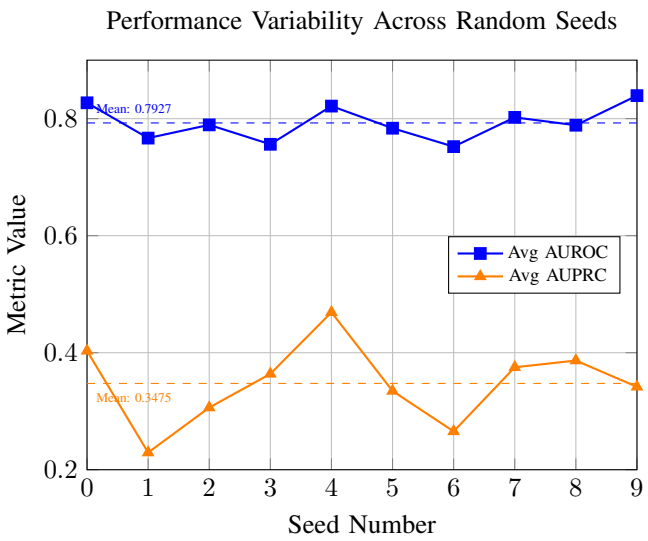

Fig. 2: Model performance per seed averaged across all dropout settings. The horizontal dashed lines represent the overall mean across all 10 seeds

All levels of dropout raised the level of ensemble AUROC and AUPRC, both peaking at a dropout of 0.1. Accuracy peaked at a dropout of 0.2.

TABLE I: Dropout Performance Comparison

Dropout	AUROC	AUPRC	Accuracy
0 (Default)	0.7895	0.3302	0.9401
0.1	0.7955	0.3543	0.9421
0.2	0.7930	0.3540	0.9439
0.3	0.7927	0.3515	0.9424

The effect of message passing depth was much less pronounced. A depth of 5, one of our ablations achieved the highest AUROC, but only by a small margin compared to the original paper's choice. The AUROC range across message passing depths was only 0.0035. However, message-passing depths of 1 and 5 both achieved a higher AUPRC than the baseline. The accuracy across models was nearly identical, only differing by 0.0012.

TABLE II: Message-Passing Depth Performance Comparison

Depth	AUROC	AUPRC	Accuracy
1	0.7867	0.3369	0.9399
3 (Default)	0.7895	0.3302	0.9401
5	0.7902	0.3402	0.9411

Despite our promising results, particularly with dropout, we cannot immediately attribute causality to our changes. The original paper only reports on the within-ensemble standard deviation. In order to definitively say that dropout or message-passing depth *causes* these changes, we would need information about the variance across ensembles. Gathering enough data to estimate the ensemble distribution would be infeasible, as the time required to train one ensemble can easily exceed 2 hours.

C. Goal Achievement

We are very happy with our results. We clearly accomplished our base goals. The original model was reimplemented, and we ran inference on the COCONUT marine products dataset [7]. In terms of target goals, we performed ablations on message-passing depth, ensemble size, and dropout. We did not hit our stretch goals. Training all those ensembles would have required way too much time, and was out of the scope of our project. Additionally, running inference on COCONUT [7] is very slow due to how large the dataset is.

V. Challenges

We ran into a few challenges on the project that made our analysis of the model's accuracy difficult.

One challenge occurred when testing the model with the COCONUT [7] dataset. The model predicted that certain molecules would have 100% effectiveness in inhibiting the bacteria. This was represented as a 1.0 prediction rating and found for 38 out of the 502,874 molecules. To determine if there was a pattern in the 38, we discovered that all had a diazonium group involved. Molecules with this group were not part of the paper's experiment; the model had never encountered them before. Thus, all of the 1.0 predictions were out-of-distribution predictions. Additionally, we know that the model cannot be applied to all molecules and can be improved with more manual testing and a larger labeled training set.

Another challenge occurred when running with a single model. We matched the paper's 240 valid molecules. This match was a coincidence of the seed-0 split, not a true reproduction. Of our 240 priority molecules, only 148 (62%) survive the 10-model ensemble. This suggests that the paper's 240 valid molecules are not stable to model instantiation.

VI. Ethics

A. Societal Impacts

Deep-learning-assisted drug discovery has the potential to be both a force for good and a force for evil. Despite the universe containing an incalculable number of molecules, scientists only have the time and resources to test a small

fraction. Deep learning has the potential to focus scientists' efforts to where it might be most fruitful. However, the same technology that could be used to create life-saving drugs could also be used to find new chemical weapons and toxins.

B. Fairness and Bias Considerations

There is a risk of bias stemming from the researchers' selection of molecules to test for their SD1 dataset. The researchers selected 7,686 molecules to test, but did not go into much detail on their selection process. If the molecules were of similar structure and did not represent molecules that could be tested with, the efficacy of the model is put into question. Only 6.25% of molecules tested were positive, translating to less than 500 active molecules in the dataset. Thus, the model is naturally biased towards predicting inactive molecules or molecules similar to the active ones it has seen. This is further demonstrated by the high number of out-of-distribution molecules earning 1.0 scores in the COCONUT dataset.

C. Privacy Concerns

While molecule structures themselves are not proprietary, testing data often is. While SD1/SD2 and COCONUT are public, most datasets are not. Research companies spend millions of dollars researching the interaction between different molecules, and rely on these datasets to recoup their research costs. Developers must ensure that third parties cannot reverse-engineer their model to glean some of their proprietary data.

D. Environmental Concerns

Training on SD1 was extremely computationally expensive. For a single 10-model ensemble, training took around 2.5 hours. Testing multiple hyperparameter combinations or estimating the 10-model ensemble distribution would have taken days. At what point is a small increase in AUROC or AUPRC not worth the energy and water cost? Increasing the message-passing depth to 5 significantly increased training time, but did not yield any improvement.

E. Limitations and Potential for Misuse

Outside of the possibility of uncovering dangerous molecules, the model's accuracy metrics can be misleading for those uninformed about the intricacies of the data. If someone were to only train on seed 0 or 4, they would see extremely high metric scores. Additionally, without understanding the skew of the dataset, an accuracy of 94% would seem more impressive than it is. This might embolden them to spend significant resources on testing molecules with little success. The predictions or metrics should not be taken too seriously, given the small size of the dataset used for training. Researchers must run more post-processing to narrow down a list of relevant molecules.

F. Concerns Addressed in our Implementation

To avoid misleading statistics produced from a lucky seed, we re-ran all of our ablations with a 10-model ensemble. Additionally, we limited our testing scope to not waste computational resources for marginal benefit.

VII. Reflection

We thought that this critical replication of the Liu et al. 2023 [1] paper turned out very successful. We were able to evaluate on the COCONUT Dataset, which was not used in the paper. We also tested some of the claims from the paper to see if they held in our implementation, and modified the model to see if it improved performance.

The model did work as expected, just like how it was implemented in the paper. However, we expected that certain hyperparameters would improve performance. The paper's implementation had a dropout of 0.0, which we thought could be improved to force the model to learn more generalized features. The paper's implementation had a message passing depth of 3, which we thought could be increased to get a broader picture of the underlying chemistry of the molecule. However, despite our results, we cannot attribute the increase in performance to our modifications. We need more information about the variance across different ensemble trials.

It surprised us that the paper's original dropout parameter was 0. The paper did not specifically explain why the hyperparameters they had in the model gave the best performance. We expected to see that a different level of dropout could improve our accuracy, as shown in the results section. It seemed that across the ensemble, the average AUROC and Accuracy were higher with a dropout level of .1 and .2. Similarly, with message passing depth, AUROC and Accuracy increased with a message passing depth of 5. We predicted that these increases would happen with our modifications, but due to the nature of the ensemble training, we cannot determine if these modifications we made cause the performance to improve until we have more information about the variance across ensembles.

A. Evolution and Pivots

Our approach was pretty straightforward throughout the process. We just stuck to our plan from the start and were able to make a lot of progress on it.

Initially, we tested hyperparameters with just 1 ensemble on seed 0. We thought this was fine at first, but then we realized that this method was not ideal to compare to the paper's results, since they used a full ensemble of 10 runs to get their values. We decided to replicate the paper's experiments and we ran the full ensembles with different levels of dropout and message passing depth, to give us a large set of data to average across. That way we can compare those changes similarly to how the paper had it, rather than just one ensemble.

If starting over, we might have worked to find a more applicable dataset to run on. The COCONUT dataset that we chose isn't just filled with molecules that might work in the lab environment. There are a lot of molecules in that dataset that could not work in the lab setting, but the model wouldn't be able to predict that since it hasn't interacted with them before. Maybe if we found a better, more applicable lab-tested molecule dataset, we might have a better overall picture of what our model could actually do, when looking at molecules

that would be up for consideration, rather than ones that might never be up for consideration because they don't work in a lab.

B. Future Work

With more time, we would have been able to examine different drug effectiveness against other types of bacteria. Maybe with a larger dataset with different labels, we would be able to make a claim that the model can predict the effectiveness of inhibition about a whole family of bacteria rather than just one single bacterium.

If we were able to find a label dataset that had the lab results from more molecules tested against the bacteria, we would have been able to have a more complete prediction set. Ideally, we would not have any out-of-distribution predictions and could apply techniques to drug effectiveness against other harmful bacteria. However, we know that our model is flawed in this sense, because it hasn't seen many types of molecules, and thus the ideas can't be applied in a broader sense.

Our work raised the question of how much manual testing needs to be done to produce a model that can actively predict the effectiveness of inhibition with 99% accuracy. We think that the shortcomings of the paper's model's accuracy are because they didn't test enough molecules manually to give the model a stronger label set to learn from. With a larger label set, this can become possible.

C. Key Takeaways

Our biggest takeaways were the fact that the model has a fairly high accuracy in predicting if certain molecules would be effective at inhibiting the bacteria. The accuracies were all around 94% even when modifying the hyperparameter. While ideally it would be in the 99th percentile for accuracy, we know that the model is not perfect, which is exemplified by the out-of-distribution predictions.

In order to model the chemical data, we had to work with RDKit and SMILES, where the RDKit converts the chemical molecules into a graph network for the GNN to analyze. SMILES displays the chemical compound in a string, representing 2-dimensional structured molecules. This was our first introduction to those tools, and we were able to apply skills that we had learned in the course to run the model.

We gained experience working with real-world data. Sometimes it's tough to apply skills learned in a classroom setting and scripted assignments with instructions to real-world data, but this project showed us how we can take the skills we learned in the assignments in CSCI1470 and apply them to evaluating a research paper.

Working as a team, we were able to divide up the work and use our time more effectively. We were able to have different roles for the ablations to the model, which allowed each person to become an expert on how modifying a singular hyperparameter changes performance for the model. This was helpful for the presentation and allowed us to have a person to explain in depth the questions that were asked of us.

VIII. Conclusion

- **Main contributions:** We provide a faithful reimplementation of the D-MPNN antibiotic-discovery pipeline of Liu et al. [1], verification of its two headline quantitative claims, three ablation studies on architectural decisions the original paper leaves unjustified, and a novel-chemistry stress test on the 502,874-molecule COCONUT marine natural products dataset.
- **Key findings:** The paper's reported AUROC (0.792 ± 0.042) and AUPRC (0.337 ± 0.088) reproduce within seed noise at 0.7895 ± 0.0295 and 0.3302 ± 0.0638. The headline "240 priority molecules" claim is not stable to model instantiation: only 148 of our single-model 240 (62%) survive the 10-model ensemble's filter, and the ensemble itself returns 314. Ablations on dropout and message-passing depth produced effects within seed noise. Depths 1, 3, and 5 are statistically indistinguishable, and dropout in $[0.1, 0.3]$ yields only modest AUPRC gains over zero.
- **Implications:** The architectural choices Liu et al. present as deliberate appear to matter less than the size and diversity of the 480-active training signal. On the COCONUT dataset, a small number of molecules received perfect 1.0 scores despite being from chemical families (e.g. diazonium-containing compounds) absent from SD1, indicating that the model overconfidently extrapolates rather than abstaining on out-of-distribution chemistry. Generalization is bounded by the chemical diversity of SD1's 7,686 molecules, and progress in deep-learning-guided antibiotic discovery will likely come from expanding the labeled dataset rather than refining the model's architecture.
- **Final thoughts:** The 2.5 hours or less required to train one ensemble are negligible compared to the months and physical cost of the wet-lab work needed to label additional molecules, but more of this labeled data would meaningfully raise the model's ceiling. Most of the implementation decisions are made to make the most out of every piece of labeled data without overfitting. Future replication efforts in this space should report variance across full ensembles rather than only within them, and they should evaluate on chemically diverse, out-of-distribution sets to surface generalization failures that within-distribution metrics tend to hide.

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